

Michael John Yuhas

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Education

Nanyang Technological University – Singapore Doctor of Philosophy (Ph.D.), Interdisciplinary Graduate Programme, 4.10/5.00 Thesis: <i>Out-of-Distribution Detection as a Safety Monitor in Cyber-Physical Systems</i>	01/2021 – 07/2025
McGill University – Montréal, Québec Master of Engineering (M.Eng.), Electrical Engineering, 3.29/4.00 Thesis: <i>Capacity Analysis of Faster-Than-Nyquist MIMO Systems</i>	08/2013 – 05/2017
Rose-Hulman Institute of Technology – Terre Haute, Indiana Bachelor of Science (B.Sc.), Electrical Engineering, 3.49/4.00 (<i>cum laude</i>) Capstone Project: <i>Surround View Vision System</i>	08/2009 – 05/2013

Research Interests

Deep Learning in Safety Critical Cyber-Physical Systems, Real-Time Machine Learning

Work Experience

St. Mary's University – San Antonio, Texas Assistant Professor, School of Science, Engineering and Technology	08/2026 – Present
Vanderbilt University – Nashville, Tennessee Postdoctoral Scholar, Institute for Software Integrated Systems	01/2026 – 07/2026
Nanyang Technological University – Singapore Research Fellow, Energy Research Institute	02/2025 – 12/2025
Mojo Vision – Saratoga, California Senior Staff Software Engineer, Software Team	07/2020 – 12/2020
Nio – San Jose, California Senior Validation Engineer, Digital Cockpit	12/2017 – 07/2020
Apple – Cupertino, California Software QA Engineer, Apps Team	06/2015 – 12/2017

Journal Publications

1. **M. Yuhas**, A. Easwaran, “Toward State-Aware Scheduling of Machine-Learning Workloads,” *Real-Time Systems*, Jun. 2025, doi: 10.1007/s11241-025-09444-w.
2. S. Das, **M. Yuhas**, R. Koh, and A. Easwaran, “Interpretable Latent Space for Meteorological Out-of-Distribution Detection via Weak Supervision,” *ACM Transactions on Cyber-Physical Systems*, Mar. 2024, doi: 10.1145/3651224.

Conference Publications

1. **M. Yuhas**, J. Soh, N. Wong, and S. Wang, “Improving Long-Term Photovoltaic Degradation Forecasting with Hybrid Neural Networks,” in *2026 International Conference on ICT for Smart Society (ICISS)*, Bandung, Indonesia, Aug. 2026.
2. **M. Yuhas**, G. Gunter, J. P. Talusan, A. Laszka, D. Freudberg, and A. Dubey, “Computing Headway Bounds under Worst-Case Bunching in Fixed-Line Transit Systems,” in *2026 IEEE 32nd International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA)*, Qingdao, China, Aug. 2026. (Outstanding Paper Award)
3. S. Gupta, **M. Yuhas**, and A. Dubey, “DA-MCTS: Deadline-Aware Action Exploration for Safe Real-Time Planning,” in *2026 IEEE 32nd International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA)*, Qingdao, China, Aug. 2026.
4. C. Gao, Y. Gao, **M. Yuhas**, and A. Easwaran, “Fault-Tolerant Offloading Framework for Real-Time Applications in Mobile Edge Computing,” in *2026 IEEE 32nd Real-Time and Embedded Technology and Applications Symposium (RTAS)*, Saint Malo, France, May 2026.
5. **M. Yuhas**, R. K. Ahir, L. V. T. Hartono, M. D. D. Putranto, A. Easwaran, and S. H. Supangkat, “Managing Charging Induced Grid Stress and Battery Degradation in Electric Taxi Fleets,” in *2025 IEEE Innovative Smart Grid Technologies - Asia (ISGT-Asia)*, Guangzhou, China, Nov. 2025, pp. 856–863, doi: 10.1109/ISGTAsia63446.2025.11431380. (Best Application Paper Award)
6. M. Prashant, A. Easwaran, S. Das, and **M. Yuhas**, “Guaranteeing Out-Of-Distribution Detection in Deep RL via Transition Estimation,” in *The 39th Annual AAAI Conference on Artificial Intelligence*, Philadelphia, PA, USA, Feb. 2025.
7. A. Bansal, **M. Yuhas**, and A. Easwaran, “Compressing VAE-Based Out-of-Distribution Detectors for Embedded Deployment,” in *2024 IEEE 30th International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA)*, Sokcho, South Korea, Aug. 2024, pp. 37–42, doi: 10.1109/RTCSA62462.2024.00015.
8. **M. Yuhas** and A. Easwaran, “Co-Design of Out-of-Distribution Detectors for Autonomous Emergency Braking Systems,” in *2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*, Bilbao, Spain, Sep. 2023, pp. 1996–2003, doi: 10.1109/ITSC57777.2023.10421953.
9. Z. Rahiminasab, **M. Yuhas**, and A. Easwaran, “Out of Distribution Reasoning by Weakly-Supervised Disentangled Logic Variational Autoencoder,” in *2022 6th International Conference on System Reliability and Safety (ICSRS)*, Venice, Italy, Nov. 2022, pp. 169–178, doi: 10.1109/ICSRS56243.2022.10067434.
10. **M. Yuhas**, D. J. X. Ng, and A. Easwaran, “Design Methodology for Deep Out-of-Distribution Detectors in Real-Time Cyber-Physical Systems,” in *2022 IEEE 28th International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA)*, Taipei, Taiwan, Aug. 2022, pp. 180–185, doi: 10.1109/RTCSA55878.2022.00025.
11. **M. Yuhas**, Y. Feng, D. J. X. Ng, Z. Rahiminasab, and A. Easwaran, “Embedded Out-of-Distribution Detection on an Autonomous Robot Platform,” in *Proceedings of the Workshop on Design Automation for CPS and IoT*, Nashville, TN, USA, May 2021, pp. 13–18, doi: 10.1145/3445034.3460509.

12. **M. Yuhas**, Y. Feng, and J. Bajcsy. “On the Capacity of Faster-than-Nyquist MIMO Transmission with CSI at the Receiver,” in *2015 IEEE Globecom Workshops (GC Wkshps)*, San Diego, CA, USA, Dec. 2015, pp. 1–6, doi: 10.1109/GLOCOMW.2015.7414179.

Posters and Demos

1. **M. Yuhas**, J. P. Talusan, A. Laszka, D. Freudberg, and A. Dubey, “Demo: Rapid Optimization of Fixed-Line Transit Holding Policies,” in *2026 IEEE 32nd International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA)*, Qingdao, China, Aug. 2026.
2. S. Gupta, **M. Yuhas**, Q. Lei, and A. Dubey, “Demo: Deadline-Aware Real-Time Planning on F1TENTH,” in *2026 IEEE 32nd International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA)*, Qingdao, China, Aug. 2026.
3. X. Gao, A. J. Singh, G. Royyuru, **M. Yuhas**, and A. Easwaran, “CRLK: Constrained Reinforcement Learning for Lane Keeping in Autonomous Driving,” in *AAMAS-25 Demos track*, Detroit, MI, USA, May 2025, pp. 3026–3028.
4. **M. Yuhas** and A. Easwaran, “Demo Abstract: Real-Time Out-of-Distribution Detection on a Mobile Robot,” in *RTSS@Work 2022*, Houston, TX, USA, Dec. 2022, pp. 26–28.

Book Chapters

1. A. Easwaran, **M. Yuhas**, S. Ramanathan, and A. Samaddar, “Real-Time Scheduling for Computing Architectures,” in *Handbook of Computer Architecture*, A. Chattopadhyay, Ed., Singapore: Springer Singapore, 2024, pp. 1–44, doi: 10.1007/978-981-15-6401-7_5-1.

Awards

Singapore International Graduate Award (SINGA)

01/2021 – 12/2024

Professional Affiliations

Member, IEEE

08/2023 – Present

Service

Reviewer, <i>Innov. Smart Grid Technol. Asia</i> (IEEE ISGT Asia)	2025
Reviewer, <i>Trans. Cyber-Physical Syst.</i> (ACM TCPS)	2026
Program Committee Member, <i>AAAI Conf. Artificial Intelligence</i> (AAAI)	2025, 2026
Program Committee Member, <i>Int. Conf. Embedded Real-Time Comput. Syst. Appl.</i> (RTCSA)	2025, 2026 [†]
Program Committee Member, <i>Symp. Appl. Comput.</i> (ACM SAC)	2027

[†]Outstanding Reviewer Award

Teaching Experience

St. Mary’s University – San Antonio, Texas

Assistant Professor, School of Science, Engineering and Technology

- Fundamentals of Logic Design (EG2313) 2026-S1
- Logic Design Laboratory (EG2113) 2026-S1

Nanyang Technological University – Singapore

Teaching Assistant, School of Computer Science and Engineering

• Digital Image Processing (CE7491)	2022-S1
• Advanced Computer Architecture (CE3001)	2021-S1, 2022-S1, 2022-S2
• Microprocessor System Design and Development (CE2107)	2023-S1
• Computer Organization and Architecture (CE1006)	2021-S2
• F1Tenth (NTU hosted MOOC)	2021-S1

McGill University – Montreal, Quebec

Teaching Assistant, Department of Electrical and Computer Engineering

• Signal Processing Hardware (ECSE436)	2014-S1
• Embedded Systems (ECSE421)	2014-S2
• Introduction to Computer Engineering (ECSE221)	2013-S2
• Electric Circuits I (ECSE200)	2014-S2

Teaching Workshops

LaTeX for Beginners Workshop, Nanyang Technological University	04/2026
Deep Learning Bootcamp III: Deep Learning for Images, Nanyang Technological University	02/2025
Data Visualization for Research Workshop, Nanyang Technological University	12/2025
Deep Learning Bootcamp: Hackathon, Nanyang Technological University	03/2024
Github Workshop, Nanyang Technological University	03/2024
Deep Learning Bootcamp III: Deep Learning for Images, Nanyang Technological University	02/2024